EXPERIMENT : 10

AIM:

Experiment to understand array function in Data science.

DESCRIPTION:

Understand array function using Numpy library.

CODE:

```
import numpy as np

# Create a random array with 9 elements
array = np.random.randint(1, 100, 9)
print("Original array:")
print(array)

# Square root of each element
print("Square root:")
print(np.sqrt(array))

# Number of dimensions
print("Dimensions:", array.ndim)

# Reshape into a 3x3 matrix
new_array = array.reshape(3, 3)
print("3x3 matrix:")
print(new_array)

print("Dimensions:", new_array.ndim)

# Convert matrix into a 1D array
print("Ravel:")
```

```
print(new_array.ravel())

# Reshape into a 3x3 matrix again
newm = new_array.reshape(3, 3)
print("New matrix:")
print(newm)

# Access row 3, columns 2 and 3
print("newm[2, 1:3]:")
print(newm[2, 1:3])

# Access row 2, columns 2 and 3
print("newm[1:2, 1:3]:")
print(newm[1:2, 1:3])

# Select rows 1 to 3, but no columns
print("new_array[0:3, 0:0]:")
print(new_array[0:3, 0:0])

# Select rows 1 and 2, column 1
print("new_array[0:2, 0:1]:")
print(new_array[0:2, 0:1])

# Select all 3 rows, column 1
print("new_array[0:3, 0:1]:")
print(new_array[0:3, 0:1])

# Select rows 2 and 3, all columns
print("new_array[1:3]:")
print(new_array[1:3])
```

OUTPUT:

```
Original array:
[67 28 67 90 96 88 30 98 54]
Square root:
[8.18535277 5.29150262 8.18535277 9.48683298 9.79795897 9.38083152
 5.47722558 9.89949494 7.34846923]
Dimensions: 1
3x3 matrix:
[[67 28 67]
 [90 96 88]
 [30 98 54]]
Dimensions: 2
Ravel:
[67 28 67 90 96 88 30 98 54]
New matrix:
[[67 28 67]
 [90 96 88]
 [30 98 54]]
newm[2, 1:3]:
[98 54]
newm[1:2, 1:3]:
[[96 88]]
new_array[0:3, 0:0]:
[]
new_array[0:2, 0:1]:
[[67]
 [90]]
new_array[0:3, 0:1]:
[[67]
 [90]
 [30]]
new_array[1:3]:
[[90 96 88]
 [30 98 54]]
```

RESULT:

NumPy array functions were successfully implemented to perform numerical operations on the given data. Various array operations were performed efficiently, helping simplify mathematical calculations and data analysis.